Parvinder Singh

Data Engineer

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Summary

Data Engineer with 4+ years of experience designing and scaling cloud-based data pipelines and lakehouse architectures on AWS using Databricks and Snowflake. Proven track record of building high-volume batch and real-time ETL/ELT solutions for healthcare and finance domains, improving data quality by 30% and enabling reliable analytics at scale. Experienced in analytics engineering, data governance, and cost-optimized data platforms supporting mission-critical business systems.

Skills

- **Databases:** Snowflake, Amazon S3, Redshift
- **Orchestration & ELT:** Airflow, dbt
- **Streaming & Processing:** Apache Spark, Kafka, Delta Lake
- **Cloud:** AWS, Azure
- **DevOps & CI/CD:** Terraform, Docker, Jenkins, GitHub Actions
- **Data Quality &** Governance: Great Expectations, Collibra, Alation
- **Visualization:** Power BI, Tableau, Looker
- **Monitoring:** CloudWatch, Grafana
- **Programming:** Python, SQL, Scala

Experience

Humana, USA | Data Engineer | Feb 2024 - Present

- Developed a data lakehouse platform on AWS and Databricks to centralize 10M+ healthcare records, increasing data reliability and accelerating analytics delivery by 30%.
- Designed and maintained data ingestion frameworks handling 100M+ rows/day, enabling analytics teams to scale dashboards for 200+ users.
- Built a data validation framework using Great Expectations integrated with Airflow DAGs, reducing production data errors by 30% and optimizing SLA adherence by 25%.
- Ensured HIPAA-compliant data handling, access controls, and auditability across healthcare data pipelines.
- Optimized Snowflake queries and warehouse costs, reducing monthly spend by 30% while accelerating BI report generation for executives.
- Designed highly available, fault-tolerant AWS-based data platforms supporting large-scale analytics workloads.
- Built and deployed Spark-based ETL workflows using PySpark, streamlining the migration of medical data from legacy systems to AWS-based data lake, Boosted data processing efficiency by 40%.

Hexaware Technologies, India | Data Engineer | Sep 2018 - Jul 2021

- Supported deployment of machine learning models on AWS SageMaker by building scalable data pipelines for feature generation and model inference, contributing to improved customer retention outcomes.
- Orchestrated ETL workflows with Airflow and CI/CD pipelines using GitHub Actions + Terraform, cutting release cycle time from weekly to daily.
- Designed scalable, fault-tolerant data architectures aligned with high-availability and low-latency requirements.
- Automated lineage and metadata management with Collibra and Airflow DAGs, ensuring end-to-end data traceability.
- Developed and optimized large-scale data pipelines on AWS, enhancing the ingestion and transformation of clinical data such as EHR and claims data, reducing ETL execution time by 30% using S3 and EMR for efficient data processing.
- Partnered with business stakeholders and analytics teams to translate requirements into scalable data solutions.
- Designed analytics engineering workflows using dbt (tests, snapshots, documentation), improving model reliability and stakeholder trust.
- Monitored and fine-tuned Spark-based jobs using AWS CloudWatch, optimizing resource allocation and cutting infrastructure costs by 20%, ensuring seamless data flow for clinical decision support systems.

Education

Master of Science in Computer Science, University of Texas at Dallas — **Aug 2021 – Dec 2023**

Bachelor of Engineering in Computer Engineering, University of Pune — **Aug 2016 – May 2020**

Projects

- Built end-to-end data pipelines using Apache Spark, AWS S3, and Snowflake as part of MS coursework.
- Processed and transformed large-scale datasets using Python and SQL, enabling analytical reporting and performance optimization.